



ATTENTION IS NOT EXPLANATION

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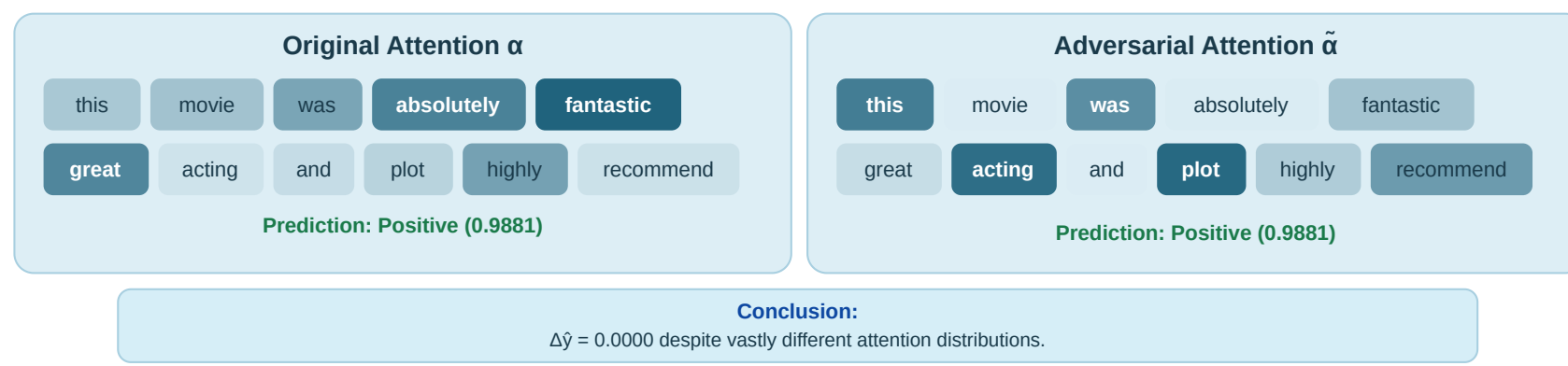
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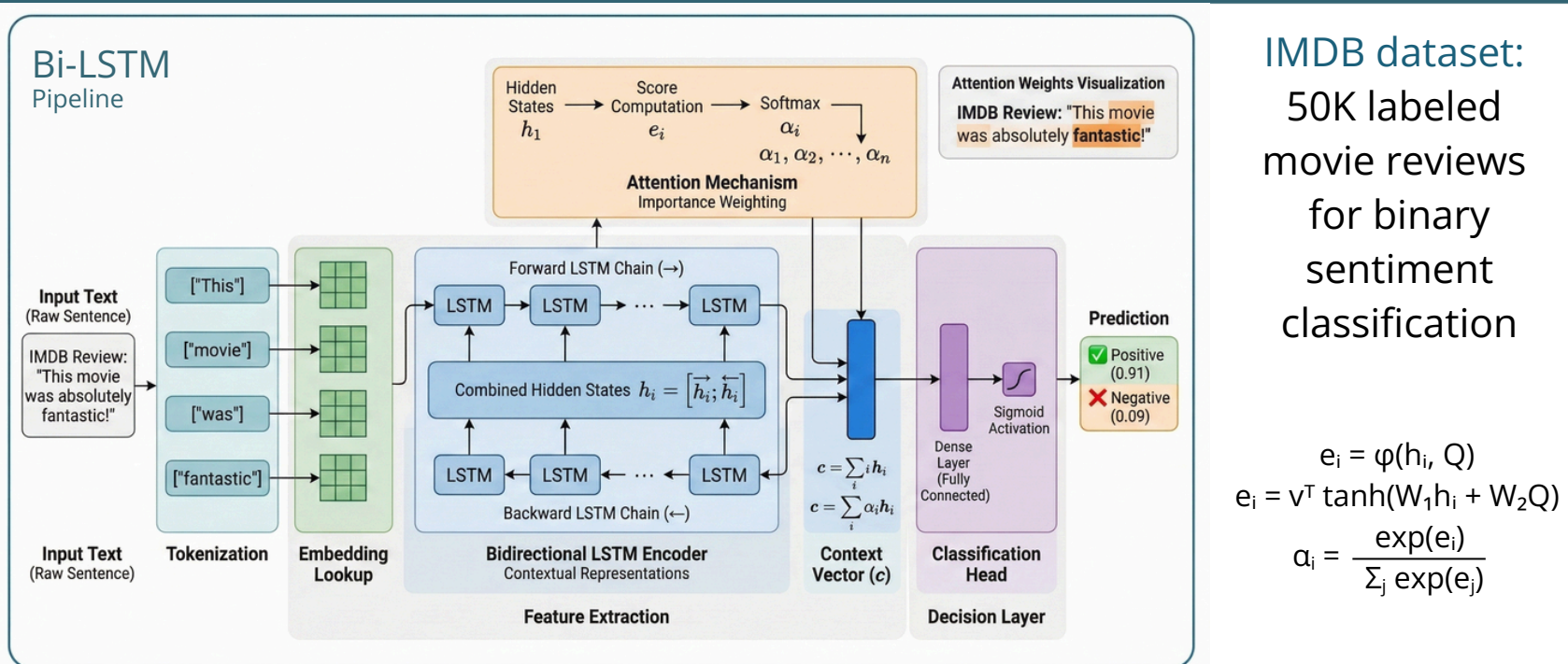


Problem Statement

- The Assumption** : Neural NLP models widely use attention mechanisms, assuming they naturally provide transparency and interpretability.
- The Goal** : Rigorously audit attention weights to see if attention give faithful explanation of the prediction.
- The Conclusion** : We demonstrate a critical interpretability gap : **Attention is not Explanation.**

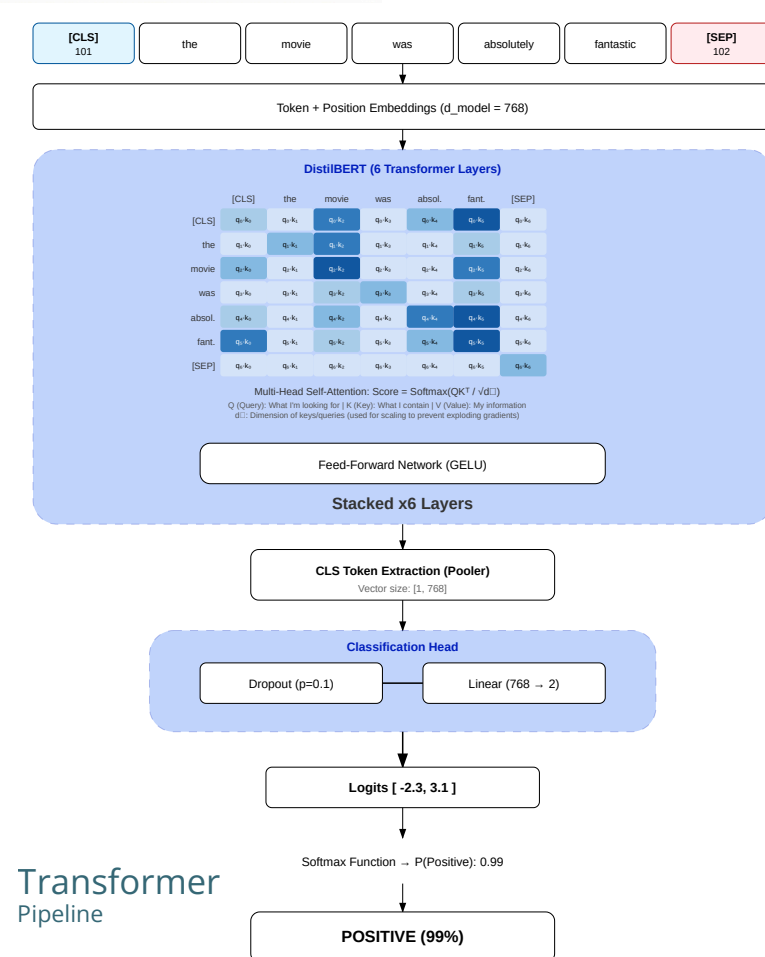


Methodology and Model Architecture



The BiLSTM produces contextual hidden states that encode the input sequence and serve as the main source for prediction. The attention mechanism assigns weights over these states to form a context vector, but does not alter the representations. Thus, predictions are primarily driven by hidden states, while attention only reweights their contribution.

The Transformer (DistilBERT) processes all tokens simultaneously through 6 stacked layers of multi-head self-attention and feed-forward networks. Each token's representation is refined by attending to every other token in the sequence. The final [CLS] token embedding captures the full sentence meaning and is passed through a classification head to produce the sentiment prediction.



IMDB dataset: 50K labeled movie reviews for binary sentiment classification

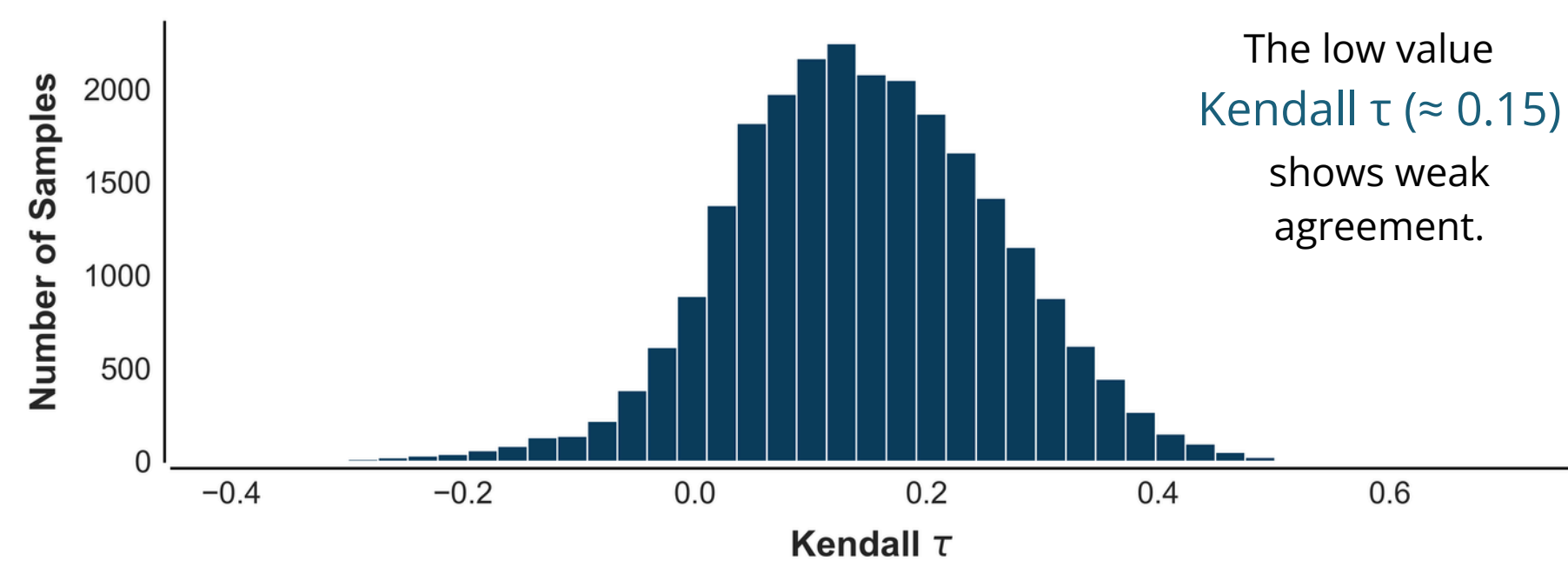
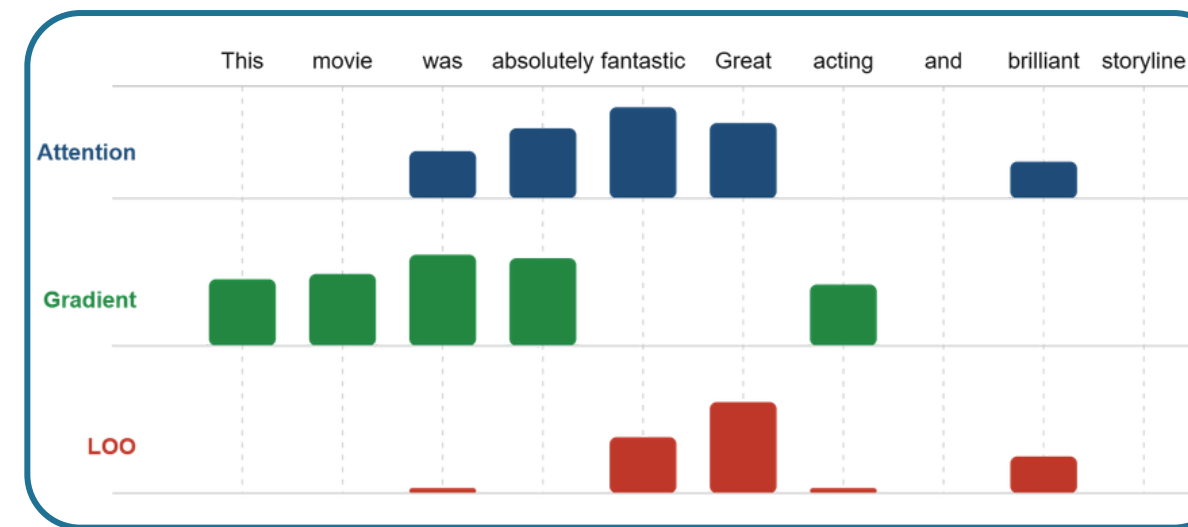
$$e_i = \varphi(h_i, Q)$$
$$e_i = v^T \tanh(W_1 h_i + W_2 Q)$$
$$\alpha_i = \frac{\exp(e_i)}{\sum_j \exp(e_j)}$$

Attention vs Feature Importance

Compare attention with gradient and leave-one-out importance.

Gradient importance: Measures how much the prediction changes for a small change in each word.

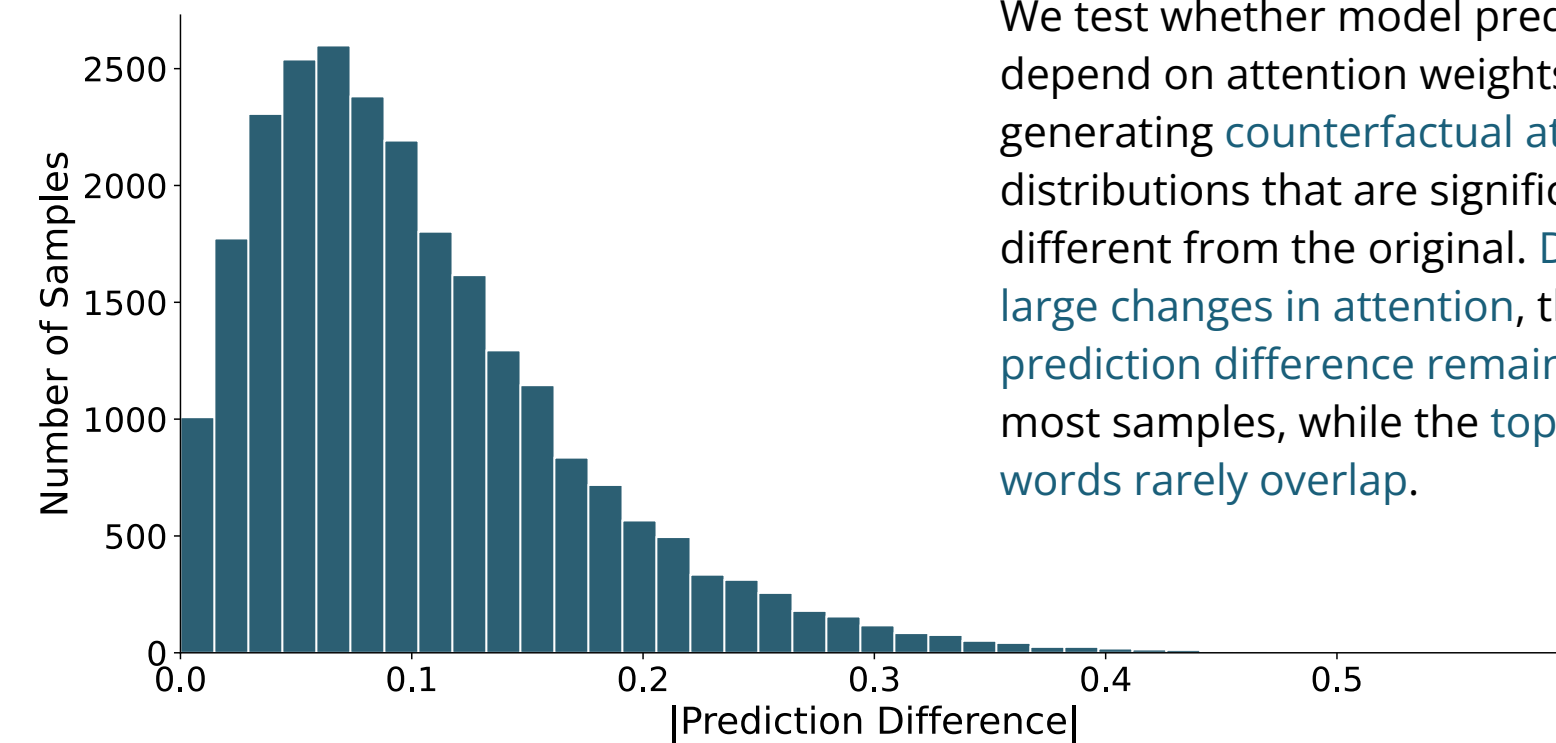
Leave-One-Out (LOO): Measures how much the prediction changes when a word is removed.



The low value Kendall τ (≈ 0.15) shows weak agreement.

Adversarial Attention

Predictions remain stable despite changes in attention



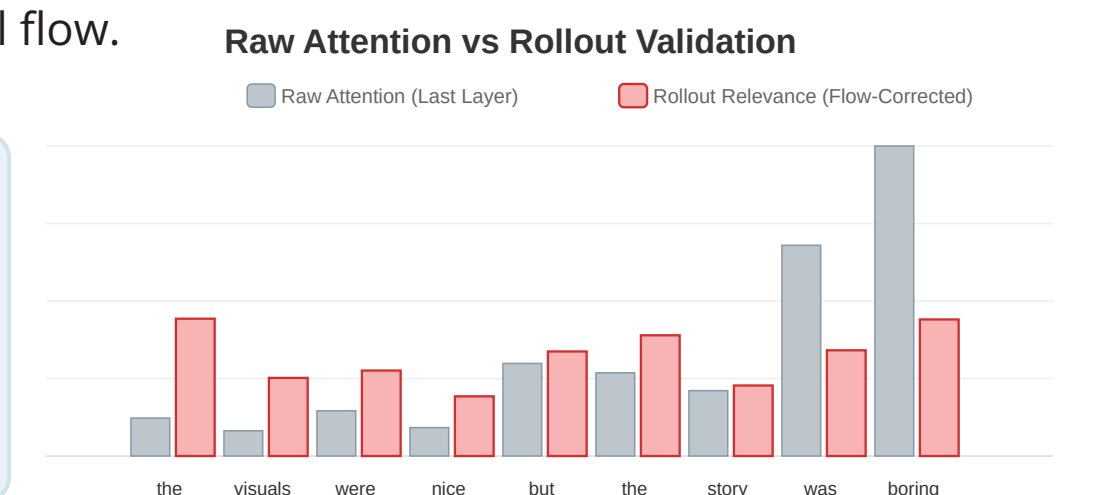
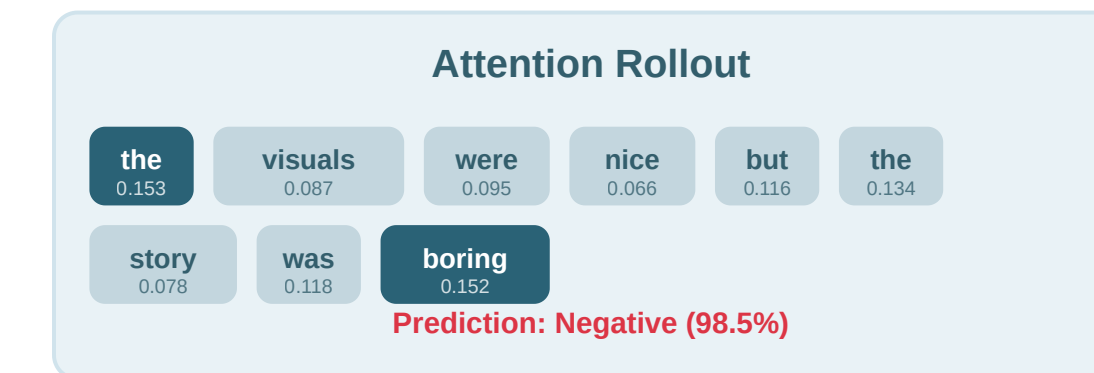
We test whether model predictions depend on attention weights by generating counterfactual attention distributions that are significantly different from the original. Despite large changes in attention, the prediction difference remains small for most samples, while the top attended words rarely overlap.

Beyond Attention: Transformer Interpretability

- The Flaw in Raw Attention** : In Transformer architectures like DistilBERT, token representations are heavily mixed via residual skip connections. Standard attention matrices completely ignore this residual flow, leading to severe misattribution of feature importance across deeper layers.

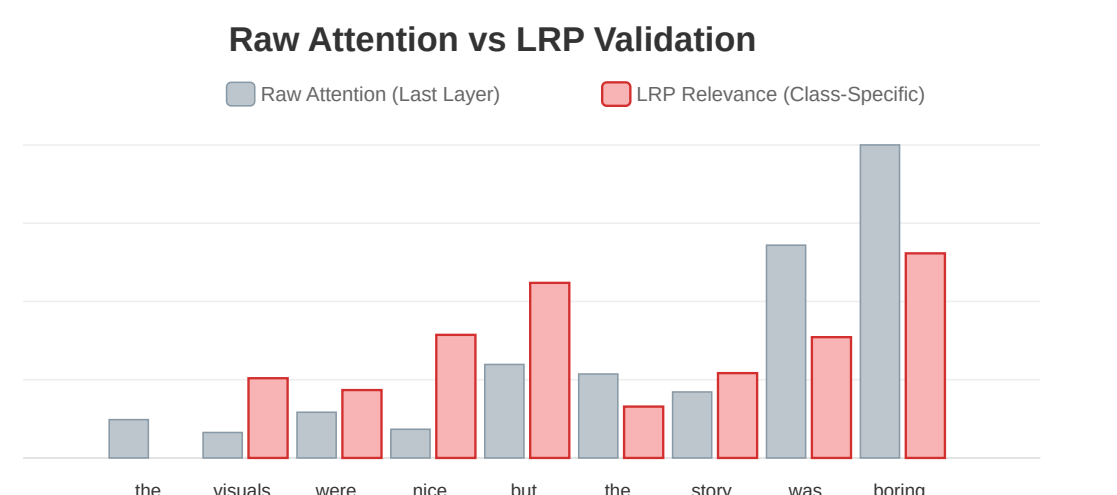
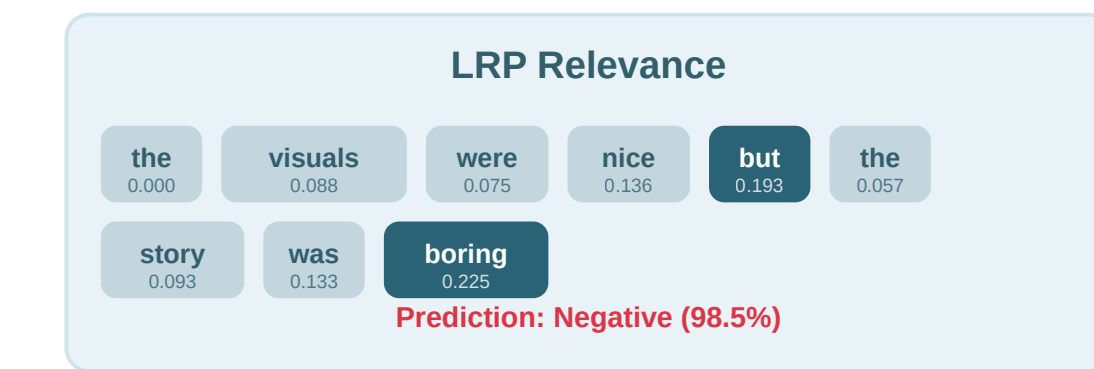
- Attention Rollout** : To compute a token's true holistic contribution, Rollout recursively tracks attention through the network while factoring in the unaltered residual flow.

$$A_{total} = (0.5A_{layer} + 0.5I)A_{previous}$$

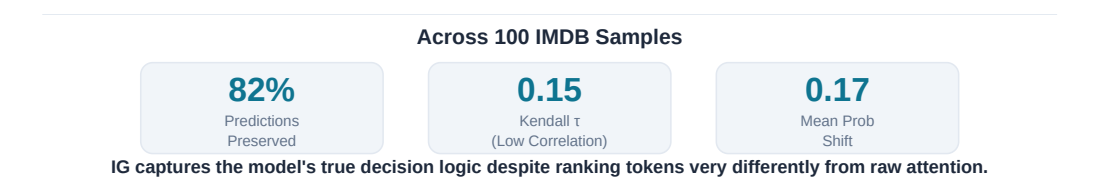
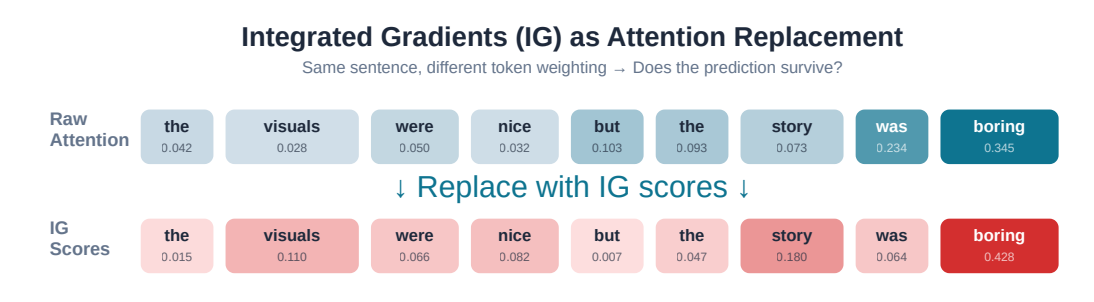
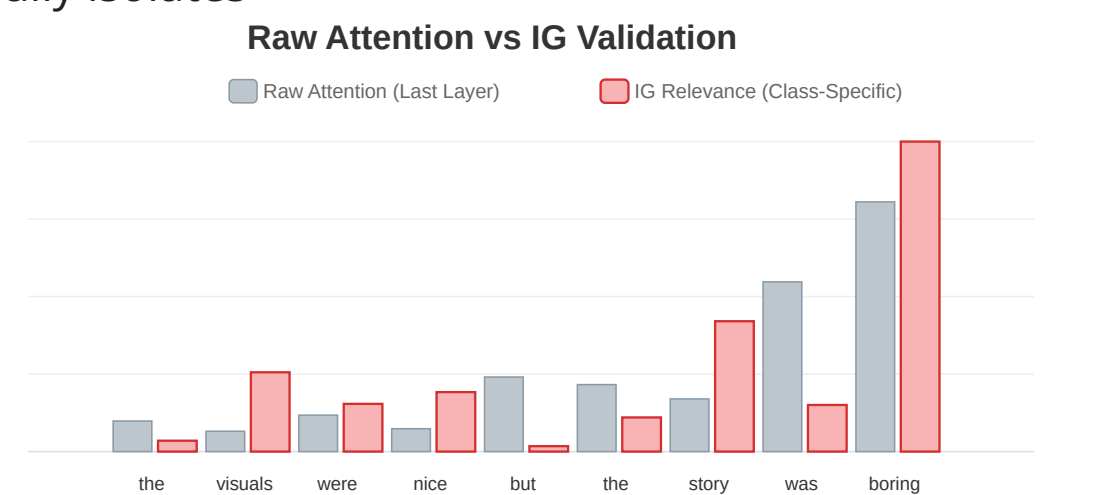
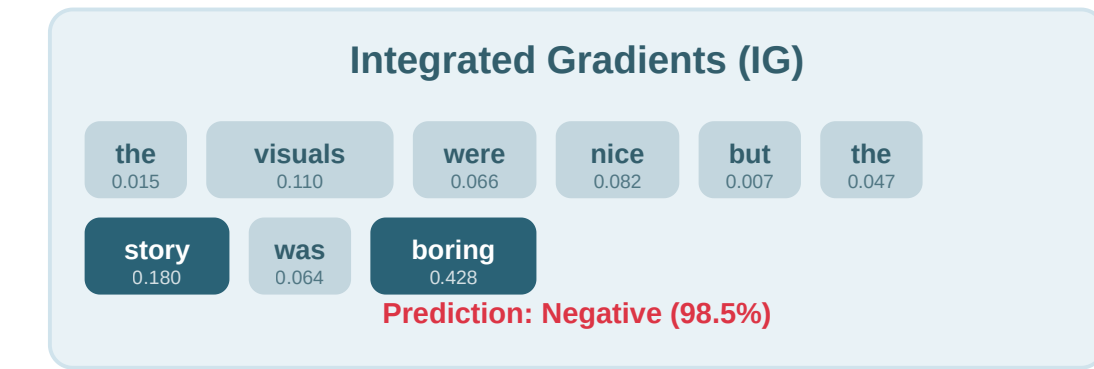


- LRP Relevance** : LRP isolates class-discriminative features by weighting the attention map by the gradients backpropagated specifically from the target class output

$$\bar{A} = \text{mean}_h(\max(0, \nabla A \odot A))$$



- Integrated Gradients** : IG interpolates word embeddings from a zero baseline up to 100% text. By bypassing internal routing artifacts that force raw attention and LRP to overweight structural stopwords (e.g., "the"), IG faithfully isolates true sentiment drivers (e.g., "boring").



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